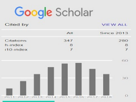
Publications

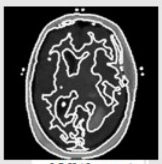


Score: 13.63
Total Reads: 3000

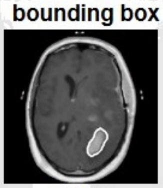
Conferences
- MICCAI
- ECML
- ACCV
- ICIP

Journals
- Pattern Recognition
- IEEE Signal Processing Letters
- Americal Journal of Neuroradiology
- Computerized Medical Imaging and Graphics

FBB



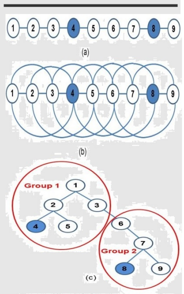
Without bounding box



With bounding box

MNI Space

MST



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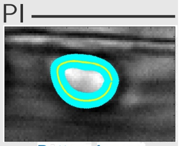
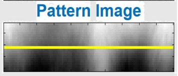
PI



Pattern Image



Research Statement

Research Area: Artificial Intelligence, Computer Vision, Machine Learning, Robotics, Image Processing, Bioinformatics, Optimization, Software Engineering, Computational Cognitive Science, Quality Control and Reliability, Industria 4.0.

Research Accomplishment: Highlights of my research accomplishments are : (i) I published thirty five (35) peer-reviewed research articles till date (October 2018). Among them, eighteen (18) papers are published on Computer Vision and Image Processing, five papers (5) on automatizing manufacturing operations by industrial robots using artificial intelligence algorithms, eight papers on welding sequence optimization, three papers on computational cognitive science, and one paper on theoretical machine learning; (ii) My research articles are published in the reputed journals like Pattern Recognition (Impact Factor (I. F.) 3.453), IEEE Signal Processing Letters (I. F. 2.528), American Journal of Neuroradiology (I. F. 3.550), and premier conferences like MICCAI (Acceptance Rate (A. R.) ~32%, rank 1 conference in computer science) , ECML (A. R. ~30%) , ACCV (A. R. ~29%), and ICIP (A. R. ~45%); (iii) I received more than 350 google scholar citations and 3000 research gate reads till date; and (iv) Recently I received a very competitve basic science research grant as young investigator and MITACS industrial internship awards for three times during my graduate studies.

Research Contributions: My research interest lies at the intersection of advancement and utilization of algorithms at various applications that can do both. My interest includes : (i) developing computationally viable solutions to large scale (complex) medical data; (ii) performing completely automated medical data analysis; (iii) offering solutions to vision based medical applications that involve multidisciplinary research (Artificial Intelligence, Machine Learning, Computational Intelligence, Computer Vision, Medical Imaging, Bioinformatics, Statistics, Reliability, Computational Cognitive Science and Software Engineering); (iv) automatizing and optimizing multifaceted cutting-edge industrial applications such as medical and bio-medical engineering, welding, industrial robots, manufacturing, transportation, and oil sand mining. My diverse educational background (Bachelor of Mechanical Engineering, Master in Quality, Reliability and Operations Research (QROR) as well as Master and PhD in Computer Science) facilitates me to conduct research in a broad range of interdisciplinary areas. Highlights of my research contributions are mentioned below.

Computer Vision, and Image Processing

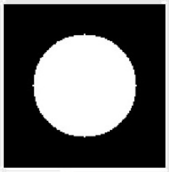
I developed several novel image segmentation, and tracking algorithms. Very few of them are mentioned below.

Brain Tumor and Edema Localization from MRI: Currently, many clinical centers maintain large archival databases of Magnetic Resonance (MR) images of brain tumors for various purposes, as this information may help physicians and radiologists to diagnose and treat new patients - e. g., by determining the effectiveness of various treatments on previous patients with similar tumor or edema volumes. As these archived MR image databases may contain a large number of studies, it is important to index their contents effectively. If these tumors were segmented, the databases could be indexed based on various aspects of the tumor, such as its size, location and type. However, accurate tumor boundary delineation is a very complex task to be accomplished accurately. We have developed an automatic, fast, and approximate segmentation technique that deals with the problem associated with accurate tumor segmentation, by locating a "bounding box" - i.e., axis-parallel rectangles, around the tumor or edema on a magnetic resonance slice exploiting the left-right symmetry of human brain [1]. We can then use this bounding box to determine tumor position, type and size to retrieve historical cases relevant to the current patient and help formulate treatment plans using deep learning. Once these bounding boxes are found on all input slices of a patient's 3-D MRI, an unsupervised mean-shift clustering (MSC) uses the locations (centroids) of these bounding boxes to separate relevant slices (slices having tumor or edema) from normal. Our papers received nearly one hundred and fifty google scholar citations till date (October 2018).

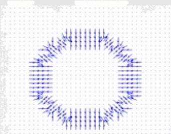
The resulting method is very fast, robust and reliable for indexing tumor or edema images for both archival and retrieval purposes and it can be used as a vehicle for further clinical investigations. This method is encoded as anomaly detection problem and can also be used to assess Estrogen Receptor (ER), Progesterone Receptor (PR), and Ki-67 that is essential in the histopathologic diagnostics of breast cancer. MSC can cluster the cancerous and non-cancerous cells of breast cancer into different stacks based on morphological characteristics of the cancerous and non-cancerous cells.

Optimal Sequence Selection for Deformable Registration: Significant cardiac and respiratory motion of the living subject, occasional spells of defocus, drifts in the field of view, and long image sequences make the registration of *in-vivo* microscopy image sequences an onerous task. We developed and implemented a novel two-stage Minimum Spanning Tree (MST)-based clustering method for deformable registration of microscopy image sequences [2]. We first construct a MST for the input image sequence. MST mitigates the registration error propagation of time sequenced images by re-ordering the images in such a way where poor quality images appear at the end of the sequence. Then, MST is clustered into several groups based on the similarity of the images. After that, an optimal anchor image is selected automatically for each group through an iterative assessment of entropy and MSE based coarse registration error and the local deformable registration is performed within each group separately. Subsequently, coarse registration is conducted to find the global anchor image selected among the whole time sequenced images and then a deformable registration is conducted on the whole sequence. Two-stage MST-based deformable registration algorithm can incorporate larger drifts and distortions more accurately than conventional one

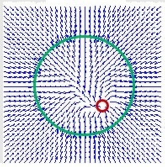
Snake



Image

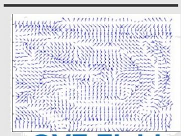


Edge Map

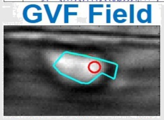


Diffusion with Dirichlet boundary condition

GVF

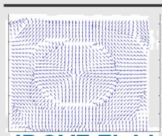


GVF Field

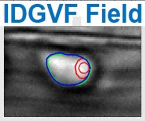


GVF Results

IDGVF

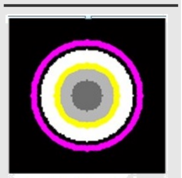


IDGVF Field

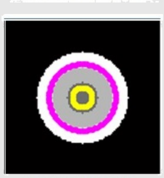


IDGVF Results

HLS



Initialization



Level Set

Hierarchical Level Set

shot registration algorithm by fine-tuning the larger amount of deformation incrementally in a couple of stages. Our method outperforms other methods on both 2D and 3D microscopy image sequences of mouse arteries used in atherosclerosis study. Our paper was published in MICCAI 2017, rank 1 conference in the area of medical imaging.

Leukocyte Segmentation and Tracking: Leukocytes play an important role in inflammation by transmigrating through endothelium and accumulating at the site of injury. Inflammation is a normal defense mechanism; however, it becomes injurious in inflammatory diseases. Inflammation research entails the study of the velocity distribution of leukocytes. Biomedical research laboratories are conducting experiments on living mice where the velocities of the leukocytes inside the veins of a mouse cremaster muscle are observed in video recordings made though a camera coupled with the intravital microscope. Leukocyte rolling velocity is an indicator of the inflammation process. To obtain the velocities of leukocyte or to initiate tracking of leukocytes, leukocytes are required to segment first. Normally skilled operators are employed to manually inspect hours of video and to extract the desired regions. Such manual processing is subject to operator errors and biases, extremely time consuming, tedious, has poor reproducibility and above all not cost effective. We developed a snake based automated segmentation technique to segment leukocytes from *in-vivo* microscopy images.

Snake is revered as state-of-the-art interactive segmentation algorithm for microscopic cell images. The introductory paper on snake has secured more than nineteen thousand google scholar citations till date (February 2018) [3]. However, snake fails to delineate object boundaries in many real-life applications which urge automation. Literature demonstrates that snake automation needs two seuqential steps: initialization and evolution. However we ameliorated snake automation by integrating a self-validation mechanism into snake segmentation framework to revolutionize snake from interactive to automatic segmentation. We fostered three novel algorithms towards snake segmentation: (a) A quadtree or Fast Bounding Box (FBB) based fast object localization method to initialize snake; (b) Interleave Directional Gradient Vector Flow (IDGVF) where novelty lies in two fronts: (i) instills Dirichlet Boundary Condition (DBC) into existing snake evolution algorithm which transmutes the gradient field anisotropic and snake is manifested as more initialization independent; (ii) computation of gradient or force field and snake evolution algorithm is performed concurrently in an interleave fashion which fortifies snake over noises; (iii) evolved snake is sieved through a claasifier such as Principal Component Analysis (PCA) or Adaptive Regularized Boosting (AR-Boost) which determines whether delineated region is a desired object and snake is revitalized as an automatic segmentation algoritgm for multiple blob-object segmentation [4]. These methods have been successfully deployed in a number of applications, namely brain tumor, edema and white matter lesion detection from MRI, leukocyte and cell detection from histopathological images, and sundries.

My future endeavor involves extending such techniques even to objects having low boundary contrast, so that automated snakes become applicable to images with poor lighting conditions (many typical industrial setups) and low-contrast cellular/molecular imagery. One of my other future works includes developing generalized object detection and tracking framework by incorporating self-validation mechanism with spatiotemporal features. One of its significant applications includes cell detection and tracking.

White Matter Lesion (WML) Segmentation from MRI: WML are thought to reflect chronic microvascular ischemic disease and often develop in persons with type 2 diabetes. Higher WML load is associated with risk factors for CardioVascular Disease (CVD), such as multiple sclerosis, diabetes, hypertension and tobacco use. Quantitative analysis of WML in large clinical trials may improve diagnosis and prognosis of the diseases affecting cerebral white matter. I have developed a novel Hierarchical Level Set (HLS) followed by Adaptive Regularized Boosting (AR-Boost) for WML segmentation. Level set is a curve evolution based image segmentation algorithm which has been manifested as a successful segmentation tool for many applications. The prelusive article of level set segmentation algorithm has bagged more than eleven thousand google scholar citations till date (February 2018) [5]. However, Level set typically creates two partitions and consequently can only capture WML of similar intensity. However, level set fails in many applications akin to WML segmentation which demands more partitions. I have developed a Hierarchical Level Set (HLS) algorithm that implements tiered iterative level sets based on residual variation to capture multiple partitions. Then AR-Boost classifies the segments into WML and non-WML classes. Thus HLS can successfully capture White Matter Lesions (WML) of various types from structural brain Magnetic Resonance Imaging (MRI) [6]. We conducted a comparative study between the CereBroVascular Disease (CBVD) with type 2 diabetes among European Americans (EA) and African Americans (AA) as a part of the NIH R01 grants (DHS-MIND and AADHS-MIND). Quantitative analysis of WML in large clinical trials can improve diagnosis and prognosis of the CBVD.

My future interest involves extending such hierarchical methods to other well-known image segmentation techniques such as Graph Cut, Locally Adaptive Thresholding and Markov Random field based segmentation method. Some of the biomedical applications may get benefit from this hierarchical method such as both cell and its interior nucleus segmentation to design automatic blood count systems. Our very recent experiment (manuscript under preparation) demonstrates that AR-Boost has high potential for multiclass classification such as for brain age prediction from pediatric brain MRI and three way-classification of Alzheimer disease (Alzheimer Disease (AD), Cognitively Normal (CN) and Mild Cognitive Impairment (MCI)). Age prediction from pediatric MRI offers better understanding of normal brain maturation as a basis for understanding atypical brain development associated with a variety of disorders and diseases.

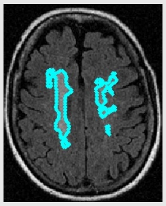
Alzheimer Disease Prediction from Structural Brain MRI: We developed a Statistical Relational Learning (SRL) based classifier for three class classification of Alzheimer disease [7]. SRL is a natural choice of model relations, it can find the relationship between the different regions of the brain representing the classes, Alzheimer Disease (AD), Cognitively Normal (CN), and Mild Cognitive Impairment (MCI). Our extensive experiments demonstrate that SRL outperforms propositional classifier such as Support Vector Machine (SVM), Decision Tree (DT) for AD classification. In future, SRL can be used for modeling, analyzing and simulating complex brain networks helping diagnosis and treatment plan for a lot of diseases and thus Artificial intelligence initiates a new horizon in personalized medicine.

Early Disease Detection from Infrared Thermography: I also had experience MITACS industrial internship training during my doctoral studies, where I worked on the project dealing with the early detection/classification of sick birds by analyzing thermal images. In the light of different food-related viruses, it is absolutely critical to have a reliable early detection method in the food supply industries/animal farm. I have developed a novel Hierarchical Variational Thresholding (HVT) followed by Regularized Regression method to delineate the warmer regions around the tympanic membrane of the chicken originated from Laryngotracheitis that is used as a non-invasive biomarker for early detection of flu. Locally adaptive variational thresholding technique applies different threshold for different pixels to handle uneven background stemming from poor or nonuniform illumination conditions which outperforms global thresholding method. However, variational thresholding typically creates two partitions and consequently can only capture regions of similar warmer temperature than background. However, variational thresholding is unable to generate more partitions in the applications akin to delineate different temperature regions over the chicken body to study the temperature distribution in the sensitive regions which is imperative for non-invasive early disease detection from thermal images. HVT implements tiered iterative variational surface based on residual variation to capture multiple partitions. Then regularized regression classifies the flocks into disease and normal classes and impedes the outbreak of avian influenza in the poultry.

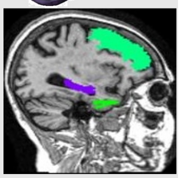
HVT can successfully capture lachrymal gland from thermal images, which is very sensitive to thermoregulatory changes associated with stress and early detection of abnormally high temperature as a result of Bovine Viral Diarrhea (BVD) can hinder outburst of the diseases in the cattle. HVT can also be used as an imperative tool for early non-invasive diagnosis of a number of human diseases from thermal images such as, peripheral vascular disorders, breast cancer, malignant melanoma, therapy response in inflammatory arthritis, visualize musculoskeletal injuries, locate tender points in fibromyalgia, diagnose complex regional pain syndrome (algodystrophy), monitor wound healing after surgery, evaluate microcirculation in vascular diseases, fever screening, thyroid disease detection, risk for diabetic peripheral neuropathy, Reynaud's phenomenon, orthopedics, and many such others.

Pulmonary Disease Detection from Proton (^{1}H) MRI: Inhaled hyperpolarized helium-3 (^{3}He) gas is a new magnetic resonance (MR) contrast agent that is being used to study lung functionality.Traditional ^{1}H Magnetic Resonance Imaging (MRI) of the lung is difficult, because the lung consists largely of air, which has a low proton density. A low proton density gives rise to a weak MR signal, but this limitation can be overcome with the use of a new class of contrast agent-hyperpolarized gas. When hyperpolarized helium-3 (H^{3}He) gas is inhaled, the airspaces fill with the gas, and the gas produces a strong MR signal, allowing a direct assessment of lung ventilation. Using (H^{3}He), high signal-to-noise ratio images of lung ventilation have been obtained. Areas of the lung that do not ventilate are deficient in ^{3}He gas and do not produce a MR signal. This lack of ventilation may be caused by airway closure as occurs in asthma or due to tissue destruction as occurs in emphysema. The unventilated areas are, therefore, seen as "defects" or dark areas on the images. Preliminary studies have shown that H^{3}He lung MRI demonstrates ventilation defects in multiple lung diseases including asthma, Chronic Obstructive Pulmonary Disease (COPD) such as, cystic fibrosis, smoking related lung disease, and bronchiolitis obliterans. Automated quantitative analysis of H^{3}He lung MRI studies can help in accurate diagnosis and prognosis of such pulmonary diseases. To evaluate the total lung ventilation from the hyperpolarized (^{3}He) MR images, it is necessary to segment the lung cavities. This is difficult to accomplish using only the hyperpolarized ^{3}He MR images, so traditional proton (^{1}H) MR images are frequently obtained concurrent with the hyperpolarized ^{3}He MR examination. Hence Proton (^{1}H) MRI is very important in several pulmonary disease clinical trials, because it helps to compute lung air intake volume that indicates the degree of progress of any pulmonary disease. Segmentation of the lung cavities from traditional proton (^{1}H) MRI is a necessary first step in the analysis of hyperpolarized ^{3}He MR images. We introduced a locally adaptive image thresholding technique that provides a smooth boundary and accurately captures the high curvature features of the lung cavities from the ^{1}H MR images. Finding the optimum values of the tuning parameters in variational thresholding is a difficult task. We have offered a novel tuning parameter free variational method using variational minimax algorithm that can automatically find lung boundary from proton MRI slices.

Our novel energy functional consists of a non-linear convex combination of an edge sensitive data fidelity term and a regularization term [8]. While the proposed data fidelity term requires the threshold surface to intersect the image surface only at places with large image gradient magnitude, the regularization term enforces smoothness in the threshold surface. To the best of our knowledge, all the previously proposed energy functional-based adaptive image thresholding algorithms rely on manually set weighting parameters to achieve a balance between the data fidelity and the regularization terms. In contrast, we use minimax principle to automatically find this weighting parameter value, as well as the threshold surface. Our conscious choice of the energy functional permits a variational formulation within the minimax principle leading to a globally optimum solution. The proposed variational minimax optimization is carried out by an iterative gradient descent with exact line search technique that we experimentally demonstrate to be computationally far more attractive than the Fibonacci search applied to find the minimax solution. Our method shows promising results to preserve edge/texture structures in



SRL



VT

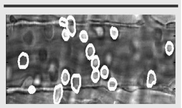
Proton MRI

Proposed

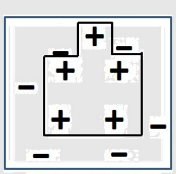
Otsu

Leuko-

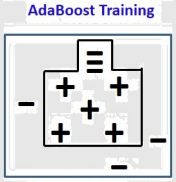
cyte



AdaBoost

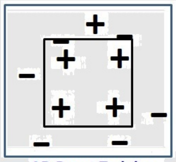


AdaBoost Training

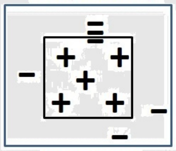


AdaBoost Test

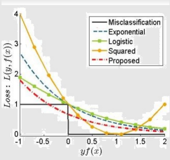
AR-Boost



AR-Boost Training



AR-Boost Test

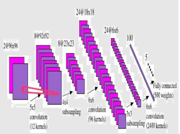


proton (^{1}H) MR images over other competing methods.

My future interest involves extending such hierarchical methods to other well-known image segmentation techniques such as Graph Cut [16], Locally Adaptive Thresholding [8] and Markov Random Field (MRF) [17] based segmentation method. Some of the biomedical applications may get benefit from this hierarchical method such as both cell and its interior nucleus segmentation to design automatic blood count systems [18]. Our very recent experiment (manuscript under preparation) demonstrates that AR-Boost has high potential for multiclass classification such as for brain age prediction from pediatric brain MRI and three way-classification of Alzheimer disease (Alzheimer Disease (AD)), Cognitively Normal (CN) and Mild Cognitive Impairment (MCI)) [13]. Age prediction from pediatric MRI offers better understanding of normal brain maturation as a basis for understanding atypical brain development associated with a variety of disorders and diseases [19].

My future endeavor involves extending such hierarchical methods to other well-known image segmentation techniques such as Graph Cut, Locally Adaptive Thresholding, and Markov Random Field (MRF) based segmentation method even for objects having low boundary contrast, so that automated segmentation algorithms would capacitate delineate object boundaries under poor lighting conditions (which is prevalent in many typical industrial setups) and from low-contrast cellular/molecular imagery. Some of the biomedical applications may get benefit from this hierarchical method such as both cell and its interior nucleus segmentation to design automatic blood count systems. One of my other future works include developing generalized object detection and tracking framework by incorporating self-validation mechanism with spatiotemporal features. One of its significant applications include cell detection and tracking. We are also intending to address few more important questions such as: can we have an unsupervised or semi-supervised classification paradigm for this self-validation mechanism for automatic segmentation? Undoubtedly, the unsupervised/semi-supervised learning can curtail the difficulties of frequent supervised learning in applications where object characteristics change considerably and frequently.

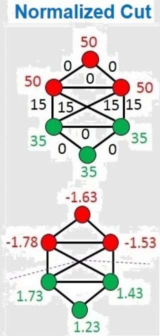
CNN



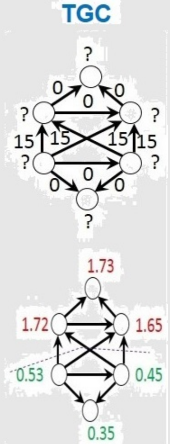
Deep Convolutional Neural Network (D-CNN): In recent years, D-CNNs have achieved great success in many problems of machine learning and computer vision due to plethora of computational power and data availability. We have implemented the idea of user interactive segmentation by random walk on a complete graph where feature vectors are learned through convnet. We developed and implemented a novel D-CNN based completely automated image segmentation alogirthm where the novelty lies in two fronts: doing random walk on a complete graph and learning feature for similarity measurement by convnet. Proposed algorithm obtains state-of-the-art performance.

TGC

Normalized Cut

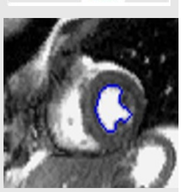


TGC



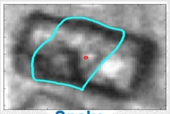
Fast Topological Graph Cut (TGC): We exploit the topological concept of cohomology to develop a fast graph cut based image segmentation algorithm which runs in linear time (O(N)). We demonstrate that for a graph, if the differential map of the function on vertices is equal to the function on edges then the function on vertices can correctly classify the vertices. Proposed algorithm first generates an oriented simplicial complex representation of an image. Next, we develop and implement a ranking algorithm which prioritizes the nodes of the weighted graph based on the similarity by solving a set of linear equations of the function on vertices in linear time. We propose a directional derivative approach to search for the function on vertices that are close to the function on weights. The differential map of the function on vertices yields the matrix of the linear matrix ranking system a simple symmetric matrix that leads the computation of the inverse of the matrix trivial and thus finds the solution of the generalized linear matrix system in linear time. Finally, we implement a standard clustering algorithm on the rank of the nodes given by the proposed ranking algorithm. We demonstrate that this proposed ranking technique has several advantages over state of the art graph cut methods: (i) the proposed method is computationally attractive; it executes in linear time; (ii) the method discovers the topological properties of the first cohomology class of the function on the weights of the graph edges that leads to an efficient graph based classifier; (iii) the proposed method performs well on super pixels; (iv) it has the ability to provide a stable segmentation under noisy conditions. Proposed algorithm obtains state-of-the-art performance.

Myocardial Border Tracking from Cardiac Magnetic Resonanc (cMR): Cardiovascular researchers are constantly developing new and innovative medical imaging technologies, striving to improve the understanding, diagnosis, and treatment of cardiovascular dysfunction. Combining these sophisticated imaging methods with advancements in image understanding via computational intelligence will continue to advance the frontier of cardiovascular medicine. cMR imaging is widely used in the study of intramyocardial function, offer a wealth of cardiac measurements, including ejection fraction, myocardial mass, myocardial wall thickening, and cardiac perfusion and directly quantifies cardiac displacement or tissue motion throughout the cardiac muscle and produce accurate spatiotemporal measurements of myocardial strain, twist, and torsion. Automated analysis of acquired tissue motion imagery through quantitative tissue tracking techniques can automatically quantify cardiac parameters, and produce results comparable to existing manual methods while alleviating researchers and clinicians from tedious, time-consuming, and highly variable manual data analyses and in turn, could potentially advance our current understanding of basic cardiovascular science.

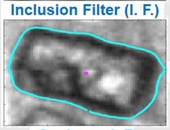


We developed a highly deformable object model for myocardial border tracking from mouse heart cine magnetic resonance imagery using level set. Automated tracking of deformable objects akin to myocardial border that change shape and size drastically is challenging. For useful results, one needs an efficient deformable object model. In this regard, we propose a novel deformable object model via joint probability density of level set function and image intensity/feature values. Given the delineated object boundary on the first image frame of a video sequence, we learn the aforementioned joint probability

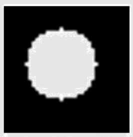
IF

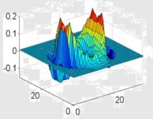

Snake

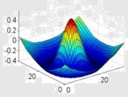

Inclusion Filter (I. F.)


Snake on I. F.

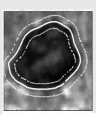
RCI

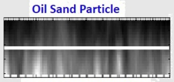

Circle Image


COIN Response Surface


RCI Response Surface

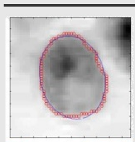
PI

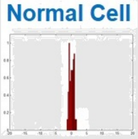

Oil Sand Particle

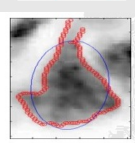

Pattern Image for object

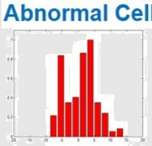

Pattern Image for Clutter

BRF


Normal Cell


Deviation from Ellipticity


Abnormal Cell


Deviation from Ellipticity

density via kernel (Parzen window) method. From the next frame onward, we match this learned probability density with the probability density on the current frame by minimizing Kullback-Leibler divergence. This minimization procedure is cast in a variational framework and a minimizer is obtained by solving a Partial Differential Equation (PDE). A stable and efficient numerical scheme is developed for solving this resulting PDE. We demonstrate the efficacy of our tracking method on myocardial border tracking from mouse heart cine magnetic resonance imagery (MRI).

Inclusion Filter (IF): We developed inclusion filter which is used to remove noises and fill the holes from the images before applying the segmentation algorithm [9]. The design of inclusion filters is motivated by self-duality. Self-duality (gray-level inversion invariance) is a desired criterion for many real-life applications, such as oil sand mining where objects appear as both darker and brighter than the background. The inclusion filter is a special class of connected filters which removes minor interior regions and clutter based on the connected component relationships defined by the adjacency forest. Connected filters offer important edge-preserving qualities which is very critical for segmentation. We demonstrated that the use of the inclusion filter significantly improves the edge fidelity and the insensitivity to initialization of active contour or snake algorithm for the oil sand and lung cavity segmentation from magnetic resonance imagery.

Automated Grading System for Breast Cancer from Histopathology Imagery: Immunohistochemical staining enables optical isolation of the epithelial structures of the breast cancer cells from normal stromal cells. Accurate assessment of morphological complexity and aberrations in the epithelial architecture are routinely used to histopathologic diagnostics and prognostics of breast cancer. Careful interpretation by a trained pathologist is not only prone to inter-observer variation, but is also tedious and time-consuming. We developed a robust coin filter based cell localization system followed by novel biologically relevant mathematical features and finally implement *k*NN regularized Support Vector Machine (SVM) to imitate the breast cancer grading system manually performed by pathologists by observing histopathological images under powerful microscope.

COnvergence INdex (COIN) filter, a successful tool for cell localization, evaluates the degree of convergence of the gradient vectors within the neighborhood (region of support) toward a pixel of interest. All previous efforts were to increase the adaptability of the region of support to make the COIN filter robust and accurate. However, improving the quality of the image gradient map was ignored, which results in poor performance of the members of the COIN family in noisy settings. We developed a new Robust Convergence Index (RCI) filter that tailors the COIN filter in a noisy environment by (a) spreading the gradient vectors within non-homogeneous object regions by convolving an Aggregated Edge Probability Map (AEPM) with an edge preserving gradient vector kernel, and (b) increasing the convergence of the gradient vectors through the integration of the sine and cosine distribution as well as the magnitude of the gradient vectors [10]. AEPM is computed through the consensus of the responses of a number of edge detectors over a wide range of scales, which lessens the effects of clutter by enforcing higher weights to the actual edges, and a non-parametric Kernel Density Estimation (KDE) is used to compute the edge probability map. RCI obtains state-of-the-art performance on histopathological breast cancer cell imagery.

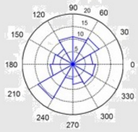
Pattern Image (PI): I developed a novel image feature that has been named as “Pattern Image" that could detect objects successfully under noisy settings. “Pattern Image" unfolds the annular band across the object boundaries to form a rectangular pattern for computational convenience that can discriminate successfully objects (leukocyte cells from *in-vivo* microscopic images used for inflammation study and oil sand particles useful for oil sand mining applications) from clutter [11]. This proposed pattern image carries texture information as well as intensity transition (dark to bight transition and vice versa) across regions of object boundaries and it outperforms other edge based features such as Gradient Inverse Coefficient Of Variation (GICOV), Average gradient strength and sundries.

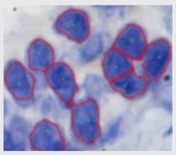
Biologically Relevant Features (BRF): We devised novel biologically relevant features useful for predicting breast cancer disease survival and recurrence from histopathological images. Towards achieving this goal, we developed three mathematical features such as (a) *Deviation from ellipticity*, (b) *Cell co-location pattern*, and (c) *Mitotic count* that translate three distinct biologically relevant features (i) *cellular pleomorphism*, (ii) *tubule formation* (randomness and abnormal aberrations of the epithelial cell structure), and (iii) *mitosis* that indicate the presence of cancer respectively. Pathologists determine the grade of the cancer of the patients based on these biologically relevant features. Our mathematical features could replicate the grading offered by pathologists that could avoid the inter- and intra-observer variability and predict the breast cancer disease survival and recurrence from histopathological images.

Imaging Genomics in Cancer Research

In recent years, a new direction in cancer research referred to as “radiogenomics or imaging genomics" has emerged to address methodologies of associating imaging features (imaging phenotype or radiophenotype) with genomic data (molecular phenotype or genotype). Recently, I received a basic science research grant for three years to find the potential association between imaging phenotype and genotype for the treatment of breast cancer. Late stage patients clearly benefit from more aggressive treatments such as chemotherapy, while for early stage patients the toxicity of the treatment may predominate the benefit. Early-stage cancers are the target of the Oncotype DX (genomic) test to find distinct subgroups within the early-stage group with different recurrence probabilities that urge for chemotherapy. However, the Oncotype DX test is very expensive but histological imaging is less expensive. We study that both histopathological findings and molecular alterations that reflect the biological properties of individual primary breast carcinomas and they demonstrate strong association among

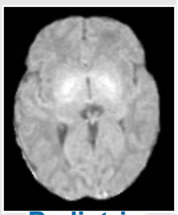
Angular Histogram



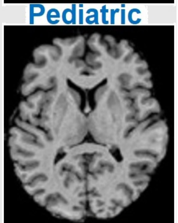


Brain

Classifcation

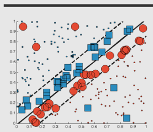


Pediatric

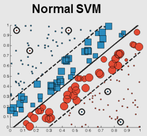


Adult

SVM

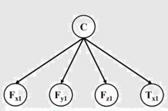


Normal SVM

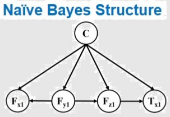


Proposed KNN-SVM

BN

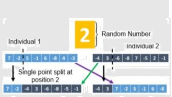


Naïve Bayes Structure



TAN Structure

GA



them. Prognostic decisions with only histological imaging for early stage patients could reduce the treament cost significantly. We advocate Deep CNN to delve the potential associations among imaging and genomic biomarkers.

Machine Learning, Soft Computing, and Big Data

A plethora of emerging Big Data applications such as, social media analysis, video surveillance, network security monitoring, personalized and precision medicine, genomics, neuroscience, economics, finance, and such others, usher processing and analyzing streams of data to extract valuable information in real-time. Big Data hold great promises for discovering subtle population patterns and heterogeneities that are not possible with small-scale data. On the other hand, the massive sample size and high dimensionality of Big Data introduce unique computational and statistical challenges: (i) high dimensionality brings noise accumulation, spurious correlations and incidental homogeneity; (ii) high dimensionality combined with large sample size creates issues such as heavy computational cost and algorithmic instability; (iii) the massive samples in Big Data are typically aggregated from multiple sources at different time points using different technologies. This creates issues of heterogeneity, experimental variations and statistical biases, and requires us to develop more adaptive and robust procedures. These challenges are distinguished and require new computational and statistical paradigm. In terms of statistical accuracy, dimension reduction and variable selection play pivotal roles in analyzing high-dimensional data. This motivates advancement of new regularization methods and sure independence screening. In terms of computational efficiency, Big Data motivate the development of new computational infrastructure and data-storage methods. Such a paradigm change has led to significant progresses on developments of fast algorithms that are scalable to massive data with high dimensionality. My research interests lie in capacitating classification algorithms for handling Big Data by incorporating data-driven regularization strategy which improves classification accuracy by avoiding overfitting and expediting the convergence. Details are given below.

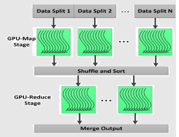
Pediatric Brain Age Prediction from MRI: Age prediction from pediatric MRI offers better understanding of normal brain maturation as a basis for understanding atypical brain development associated with a variety of disorders and diseases. AR-Boost has high potential for multiclass classification such as for brain age prediction from pediatric brain MRI. Adaboost is one of the state-of-the-art classifers for two class classification. I developed a novel Adaptive Regularized Boosting (AR-Boost) by incorporating a regularization parameter into the boosting loss function that relaxes the *weak learner assumption* of boosting algorithm and thus facilitates a natural extension of boostging algorithm for multiclass setting [12]. I advocated a regularized exponential loss function for AR-Boost which enforces more weight than Adaboost on misclassified samples at each iteration to classify correctly in the immediate iteration and consequently leads to early convergence. Unlike Adaboost, the user can select optimal weights of the features through cross-validation. Currently, I am deploying AR-Boost for pediatric brain age classification from MRI which can help in early non-invasive diagnosis and prognosis of abnormal pediatric brain development.

kNN Regularized Support Vector Machine (kNN-SVM): We describe a novel approach to combining the properties of two well-known formalisms: k-Nearest Neighbors (kNN) and Support Vector Machines (SVMs). We introduce a k-Nearest Neighbor (kNN) based regularization term into SVM optimization framework which improves the classification accuracy of SVM by leveraging the strengths of both SVM and kNN. The proposed loss function minimizes the slackness and maximizes the surplus, provides weights to the slack variables based on the degree of their k-NN and thus introduces a novel regularization strategy into SVM framework. Our approach extends the total margin SVM, which considers the distance of all points from the margin; each point is weighted based on its k neighbors. The intuition is that examples that are mostly surrounded by similar neighbors, i.e., of their own class, are given more priority to minimize the influence of drastic outliers and improve generalization and robustness. Thus, our approach combines the local sensitivity of kNN with the global stability of the total-margin SVM. We analyzed its theoretical properties and empirically demonstrate improved generalization and robustness in noisy settings for both binary- and multi-class classification. We are now extending the idea of kNN-SVM in Big Data domain by ameliorating the bottlenecks of kNN in high dimensional framework.

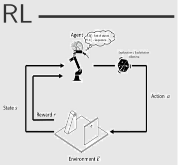
Bayesian Network (BN) Classifier: Motivated by the success of Bayesian classifier with strong assumptions of independence among features, called Naive Bayes (NB), we deployed Bayesian Network (BN) classifier with slightly less restrictive assumptions of independence among features, called Tree Augmented Naive Bayes (TAN) to classify successfully the robot execution failures associated with automation of peg in hole insertion task by Industrial Robot, Motoman MH-6 [13]. TAN approximates the interactions between attributes by using a tree structure imposed on the NB structure that generalize the NB classifier and explicitly represent statements about independence which outperforms Naive Bayes, yet at the same time maintains the computational simplicity (no search involved) and robustness that characterize NB.

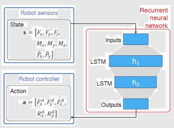
Exploiting Domain Knowledge for Early Convergence of Genetic Algorithm (GA): Genetic algorithm (GA) is a heuristic search and optimization technique inspired by natural evolution. They have been successfully applied to a wide range of real-world problems of significant complexity. Success of GA depends on smart choice of selection, crossover and mutation mechanism and fitness function as well. We advocated domain knowledge in selection, crossover and mutation algorithms that expedites the convergence and makes GA more robust for combinatorial optimization problems. We developed a novel restricted weak integer composition based sample initialization algorithm which introduce diversity in the population and in turn increases likelihood of convergence for solving progressive censoring throuh GA. In another effort, Algebraic Method (AM) is a well known analytical method for robot inverse kinematic problem. However, this method demonstrates poor convergence rate and falls into local minima in many occasions. We developed a GA based solution where AM has been introduced to reduce the search space and thus helps in speeding up the convergence rate for GA. Like Differential

BigData

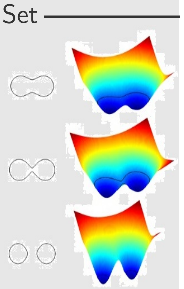


RL

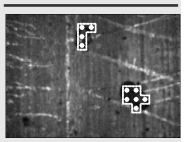


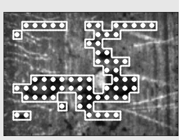


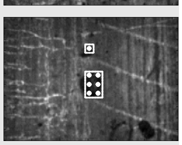
Level Set

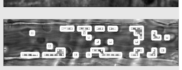


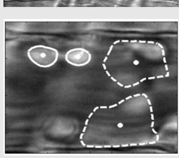
Quad Tree

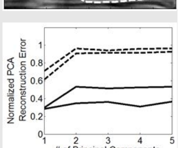


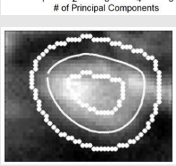


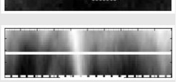












Evolution (DE), we introduced a novel vector differential based crossover algorithm in both binary and continuous domains which increases the likelihood of avoiding local optima. We also advocated data driven parameter optimization method for GA [14]. We are now performing parallel and online implementation for GA to offer realtime solutions.

Tailoring Classification Algorithm for Big Data: Support Vector Machine (SVM) is currently the state-of-the-art model for many classification problems but they suffer from the complexity of their training algorithm which is at least quadratic with respect to the number of examples. Conventional Sequential Minimization Method (SMO) is not computationally efficient for using Support Vector Machine (SVM) for large scale problems. We are conducting research on facilitating SVM for solving large scale problems using coordinate descent method. We are also introducing the idea of mixture of experts, divide and conquer, Bayesian committee machine, and MapReduce algorithms in classification algorithms such as SVM, AdaBoost, and Decision Tree for enabling them to tackle Big Data where training data is divided into a number of clusters, run training algorithm on each of the cluster separately and parallelly and then amalgamate them to generate the training model using the mixture of experts. In an alternative procedure, we are implementing a novel clustering based sampling technique to generate a representative sample of the Big Data and perform training algorithm on the sample instead of the whole large dataset to alleviate massive computation.

To Improve Medical Diagnosis through Decision Tree: In medical decision making (classification, diagnosis, etc.) there are many situations where decision must be made effectively and reliably. Conceptual simple decision making models with the possibility of automatic learning are the most appropriate for performing such tasks. Decision tree is a reliable and effective decision making technique that provide high classification accuracy with a simple representation of gathered knowledge and they have been used in different areas of medical decision making. In addition, automated decision making models developed for medical applications need to be interpretable akin to decision tree that help medical practitioners for making diagnostic and prognostic decisions for treating diseases. We are deploying decision tree for making diagnostic decisions for Non-Alcoholic Fatty Liver Disease (NAFLD). Automated decision tree can successfully diagnose NAFLD in 96% cases using only biochemical features. However, the conventional clinically-oriented metabolic feature based decision tree developed by exploiting the knowledge of the physician (domain expert) can clasify accurately only in 70% cases. We implemented lookahead decision tree that can compensate immediate wrong decision as well as incorporate induction of XOR along with conjunction and disjunction.

Improving exploration in Reinforcement Learning (RL) through Domain Knowledge: We presented a novel work on how to improve exploration in reinforcement learning using domain knowledge and knowledge-based approaches to reinforcement learning. Reinforcement learning algorithms normally face the problem of deciding whether to execute explorative or exploitative actions, and the paramount goal is to limit the number of executions of suboptimal explorative actions. We achieve this through better understanding of domain knowledge and understanding of algorithms' and domains' properties. We incorporated the knowledge of thermal affected zone and joint rigidity of the system while solving welding sequence optimization through RL [15]. At present, we are deploying monte carlo tree search for welding sequence optimization where exploration-exploitation dilemma is being balanced by UCB (Upper Confidence Bound).

Deep Reinforcement Learning (DRL): Motivated by unprecedented success of human level control through DRL [16], we are conducting research on developing a novel object detection algorithm through DRL. The agent (proposed DRL based object detection algorithm) automatically learns a policy which computes the probability of taking an action (translation and scale) in every state so that the agent can reach to the final state (desired object circumscribed by the bounding box) from the initial state (the whole image itself is circumscribed by the bounding box). DRL is adept in reducing run time complexity without compromising the detection accuracy by avoiding searches in the areas where the probability of having objects is very low. The agent constructs a deep learning feature representation of the current state (bounding box), which is then used by the Deep Q-Network (DQN) to determine the next suitable course of action so that the goal can be achieved very quickly. We are also working for developing DRL based model for robot inverse kinematic problem and peg in hole automation. Reduction of state space is one of main challenges in these problems which are being handled through DRL.

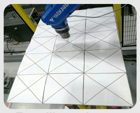
Machine Learning and Computer Vision in Manufacturing Industry

The nature of manufacturing systems faces ever more complex, dynamic and at times even chaotic behaviors. In order to being able to satisfy the demand for high-quality products in an efficient manner, it is essential to utilize all means available. One area, which saw fast pace developments in terms of not only promising results but also usability, is machine learning. The manufacturing industry today is experiencing an unforeseen plethora of available data comprising a variety of different formats, semantics, quality, e. g. sensor data from the production line, environmental data, machine tool parameters etc. Different names are used for this phenomenon, e. g., Industria 4.0, Smart Manufacturing, and Smart Factory. This increase and availability of large amounts of data is often referred to as Big Data. The availability of quality-related data offers potential to improve process and product quality sustainably, quality improvement initiatives, manufacturing cost estimation, process optimization e.g., monitoring, control, scheduling, diagnostics, troubleshooting and overall better understanding of the customer's requirements. Highlights of my contributions in maunfacturing are mentioned below.

Developemnt of Machine Learning (Data Driven) based Controller

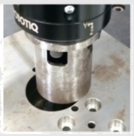
Many industrial tasks performed by robots require position-force control. In practice, engineers have to tune the force control gains on a case by case basis, in order to adapt to the different stiffness, damping or clearance of work pieces (environment).

Robot

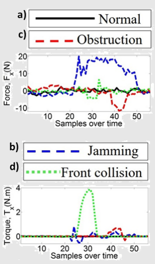


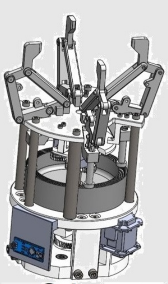
Travelling salesman

Dual arm robot

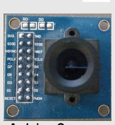


Obstruction





Gripper

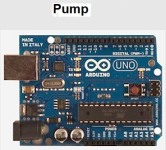


Arduino Camera

Ultrasonic sensor

Temperature sensor

Pump



Arduino Board

In contrast to the non-trivial gain tuning of conventional controller, since data-driven learning allows to reduce the amount of engineering knowledge, we advocate a different approach and speed up learning by extracting more information from data. In particular, we learn a probabilistic, non-parametric Gaussian process transition model of the system. By explicitly incorporating model uncertainty into long-term planning and controller learning our approach reduces the effect of model errors, a key problem in model-based learning. It turned out that proposed technique leads to a significant improvement over knowledge-based solutions. The reason is that the machine learning is not limited by the imagination of an expert, but is directly evaluating the process behavior instead. In addition, data driven controller outperforms in unforeseen situations.

Combining Sensors to Improve the Accuracy of the Controller

Designing Arduino based controller for controlling the level of the water in the tank using ultrasonic sesnor is a well-known problem. I with my students have incorporated multiple sensors: ultrasonic, visual sensor (camera) and temperature sensor to control the level and temperature of the water in the tank which in turn increase the accuracy of the controller. Similarly, line following by robot using infrared sensor is a popular problem. We developed a Raspberry based controller using multiple sensors: infrared, camera and ultrasonic that can deviate from the line in the presence of obstacles and again return and follow the line. We implemented camera calibration algorithm and utilized kalman filter to combine multiple sensors.

Automation of Industrial Robots for Manufacturing Operations

Automatizing Peg-Hole Insertion Task by Industrial Robot: Parts mating, peg in hole insertion or assembly operation is the most common operation in industry production, but autonomous execution by robots can significantly increase overall productivity. Peg in hole insertion task is a topic largely addressed and long standing problem in robotic research, the popularity of which is not only due to its importance in many industrial assembly tasks, but also for its complexity as a control problem that requires both position and force regulation. We developed both computer vision and without computer vision based algorithms to automate the peg in hole insertion task. We implemented a fast template matching algorithm and followed by non-linear camera calibration method to automatically finding the peg and hole in the complex working environment [17]. However, occlusion and imprecise sensor make the automation task difficult in several occasions. Hence, we portrayed the problem of peg in hole insertion search as a travelling salesman based grid search problem and we automatically look for the work pieces through force-torque sensor built in robot end-effector. Then, we developed both impedance and machine learning based controller for regulating force control. We also offered fast and accurate convergence of robot inverse kinematic solutions by incorporating domain knowledge into novel multi-objective genetic and differential evolution algorithms [18]. We executed the real experiment on Motoman MH-6, gantry or cartesian, and Crustcrawler AX-18 industrial robot. We also facilitated the use of Motoman SDA-20 dual-arm robot equipped with two three fingered gripper and a force/torque sensing capability for human like operations such as the peg-in-hole tasks by learning contact states using ANN Fuzzy ARTMAP architecture.

Design a Novel Adaptive and Reconfigurable Robot Gripper using FEM: In another effort, with one of my master students, we developed and implemented a novel Finite Element Method (FEM) based modelling for single degree-of-freedom adaptive and reconfigurable four-fingered robot gripper for automating industrial applications. FEM based deformation and stress analysis facilitates: (a) minimizing the cost of the gripper by selecting cost-effective material that can grab a wide range of objects, and (b) estimating the remaining useful life of the critical components through fatigue analysis.

Probabilistic Visual Slam: We developed a modified extended Kalman filter and particle filter algorithm as well as we exploited weighted Harris Corner and Sift based point features for visual Simultaneous Localization And Mapping (SLAM) algorithm. We evaluated the performance of our SLAM implementation from two perspectives. First, we compared the robot's estimated trajectory with respect to the ground truth. Second, we evaluated the quality of the generated map by comparing distances between landmarks suggested by the generated map with the corresponding physical distances. Our results illustrate that the estimated distances between landmarks are reasonably close to the groundtruth. Also, the relative distance estimated between selected landmarks in the map converges to a stable value which is close the ground truth.

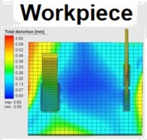
Automation and Optimization of Welding Process

Welding Sequence Optimization: Welding is the most common fabrication process which is extensively used in a wide range of industries such as automotive, shipbuilding, aerospace, pipelines, heavy and earth-moving equipment. Welding deformation and effective (residual) stress derive several negative impact to the manufacturing process and add additional cost in various ways, such as constraints in the design phase, extra operations, cost of quality and overall capital expenditure. With one of my PhD students, we developed novel modified Lowest Cost Graph Search (LCGS) [19], multi-objective Genetic Algorithm (GA) [14] as well as Reinforcement Learning (RL) [15] based Welding Sequence Optimization (WSO) techniques for reducing welding deformation and residual stress. We first demonstrated in literature the effectiveness of WSO using the the combination of conventional and Low Temperature Transformation (LTT) welding wires in the same weldment. LTT increases the fatigue life by introducing compressive residual stresses.

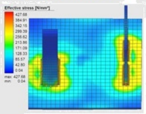
Mathematical Modelling of Weld Bead in WAAM: We developed a novel nonlinear regression followed by Taguchi method for modelling of welding bead geometry in Wire and Arc Additive Manufacturing (WAAM). WAAM has been evolved as an alternative to traditional manufacturing specially for fabricating large aerospace metal components that feature high buy-to-fly ratios. Bead modelling determines the optimum weld settings corresponding to the desired bead geometry. The strength, density and homogeneity of the deposited material depends on the bead geometry. Controlling the bead geometry

Welding

Workpiece

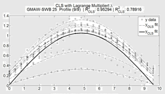


Deformation

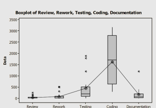


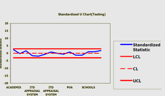
Stress

WAAM

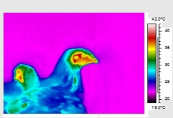


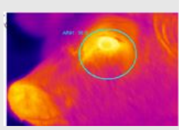
SW





IP





QC

of the workpiece is much easier and is used to predict its mechanical peroperties.

Computational Cognitive Science

As a statistician it was a matter of curiosity for me to measure mental and psychological processes. Statistical models are being applied to various disciplines outside engineering. Another thematically distinctive area of research is Applied Psychology. I felt motivated to develop statistical models, an also understand how socio-psychological processes in behavior may be predicted. This kind of research also has relevance for artificial intelligence and computational systems of perception and simulation. I participated in experiment design and analysis for three different studies: (i) color perception: we evaluate cognitive perception of the different segments of the population on the color “Rosa Mexicano", which is widely used in art, fashion, painting, architecture, and crafts [20]; (ii) statistical measurement of perception in interactive game: we designed an interactive media to check whether a short real time intervention could act as a mood changer for adolescents, in a stressful learning scenarios such as classroom and respondents' reaction were collected on a combined psychological emotional-valence scale [21]; and (iii) Computer Generated Biological Imagery (CGBI) to stimulate horripilation: we studied the specific effect of visual imagery of physical sensational horror which aims at evoking effects of chills, or goosebumps involving nervous shudder and withdrawal from objects of contemplation. Statistical tools and techniques (factor analysis, ANOVA), visualization (pie-bubble chart), and image processing algorithms have been implemented in these projects.

Industrial Research

Image Processing: I was awarded MITACS Canada industrial internship award three times during my doctoral studies, where I worked on the projects dealing with (i) the early detection/classification of sick birds from thermal images; and (ii) Automatic passenger counting from Light Railway Transit in Edmonton. In the light of different food-related viruses, it is absolutely critical to have a reliable early detection method in the food supply industries/animal farm. My responsibility was to image acquisition (mount web camera at the ceiling and grab the top views to minimize occlusion). I developed a novel variational thresholding method to delineate the warmer regions around the tympanic membrane of the chicken that is used as a non-invasive biomarker for early detection of Flu. I also implemented parametric curve to automatically detect the passenger heads, followed by optical flow based tracking algorithm and finally validation algorithm to minimize false positive.

Statistical Software Engineering: I completed several industrial projects including: (i) A graph theoretic approach to assess the quality of the class of object oriented software on the basis of a set of metrics and developing a new metric by incorporating the class dependency information; (ii) A bootstrapping method to estimate the efforts required to fix the number of bugs reported by the customer; (iii) A Bayesian approach to evaluate methodology for establishing baselines of build efforts on the basis of SAC (Simple/Average/Complex) methodology; (iv) An integer programming based optimization algorithm for software test case selection; and (v) A regression based software defect prediction methodology.

Qualty Control and Reliability: Currently I am supervising a group of students in several industrial projects that include: (i) Statistical tools and techniques (pareto analysis, nonparametric control chart and sampling plan) for visual inspection of electronic chip making process; (ii) A multiple sensor based Hiddel Markov Model (HMM) for estimating Remaining Useful Life (RUL) of the critical components of a hydraulic system; and (iii) Online removal of chattering by monitoring and detecting natural frequency for a robot guided drill system through kalman filter based control chart.

Future Research Plan

In the long term of my research career, I am interested to solve application-driven large scale (complex) image analysis. In my understanding, solving such complex real-life applications is not a simple task to design smarter and faster algorithms. I believe that alternative theoretical framework could be designed to optimize the solution to these problems. Below are two significant machine learning and image segmentation problem explaining the necessity of further research in the field.

In classification problems, Adaboost with trees is called the “best off-the-shelf classifier in the world". However, the exponential loss function for Adaboost incurs substantial misclassification error rate as the penalty increases exponentially for large increasing negative margin. To compensate this loss, we have introduced one additional term to the exponential loss function that acts as a regularizer and in turn leads to early convergence. However, the presence of even moderate amount of outlier can mar our proposed method. So, a natural question arises: is there any alternative to Adaboost that can accommodate a computationally efficient, outlier-resilient algorithm for large scale complex classification problems?

On the image segmentation, nowadays, it is a consensus among majority of image/vision experts that automated segmentation of natural images is an extremely difficult problem. Not surprisingly, so far, the success of segmentation has been demonstrated in application specific imageries, where a priori knowledge about object(s) is available. Markov Random Field (MRF) and related derivatives provide a rather limited means to incorporate a priori object information. Some recent investigations of object model based techniques have shown to remove some limitations of MRF-type limited object models. In these frameworks, an object is composed of its constituent parts in a prescribed spatial order. However, reasonably fast computations are yet to catch up with these theoretical frameworks in large complex segmentation problems. I like to investigate these issues of computations vis-a-vis the theoretical frameworks in the long run.

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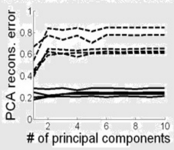
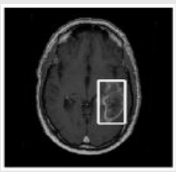
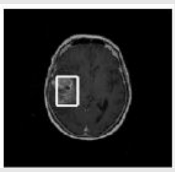
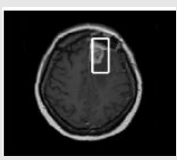
PCA



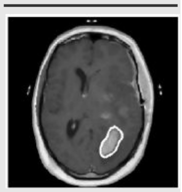
FBB

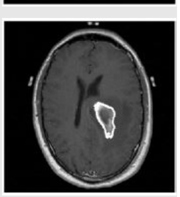


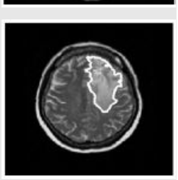


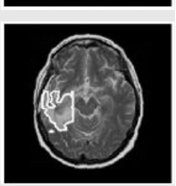


Tumor Segmentation









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